Bank Analytics Dashboard

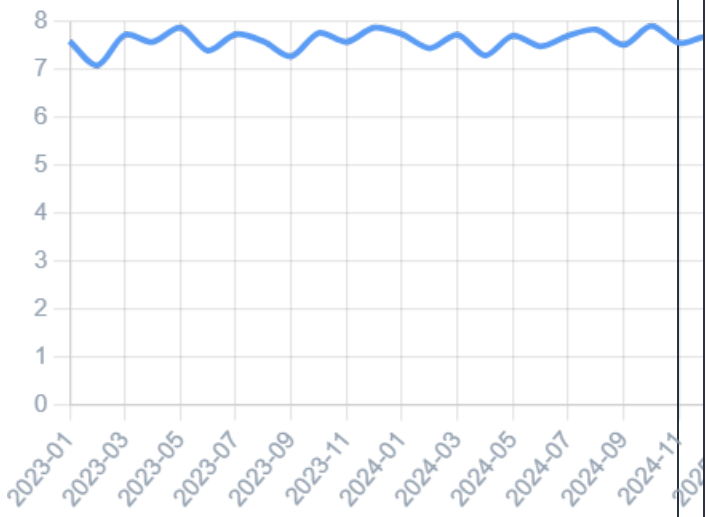
50K customers · 70K accounts · 500K transactions (2023–2025) · 20K loans · Amounts in ₹ (Cr = crore)

Customers	Accounts	Transactions	Total Balance
50,000	70,000	500,000	₹487.5 Cr

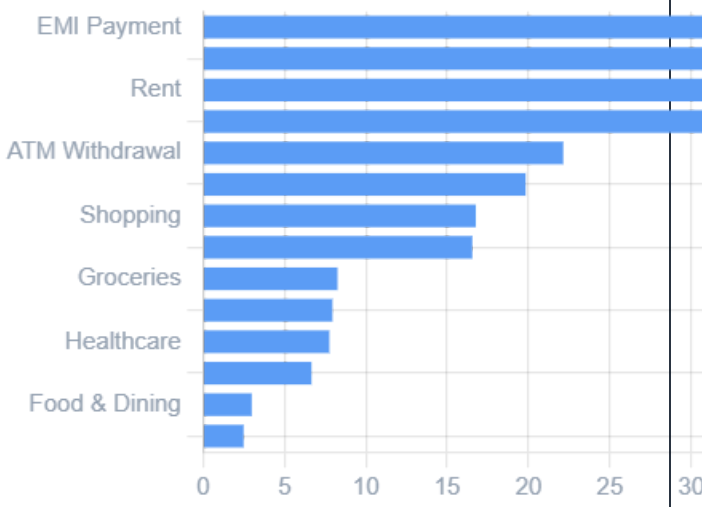
Total Spend	Loan Book	Default Rate	Fraud Rate
₹274.3 Cr	₹2,660 Cr	5.8%	0.61%

Failed Txns	Active Customers
3.54%	88.0%

Monthly Spend (₹ Cr)



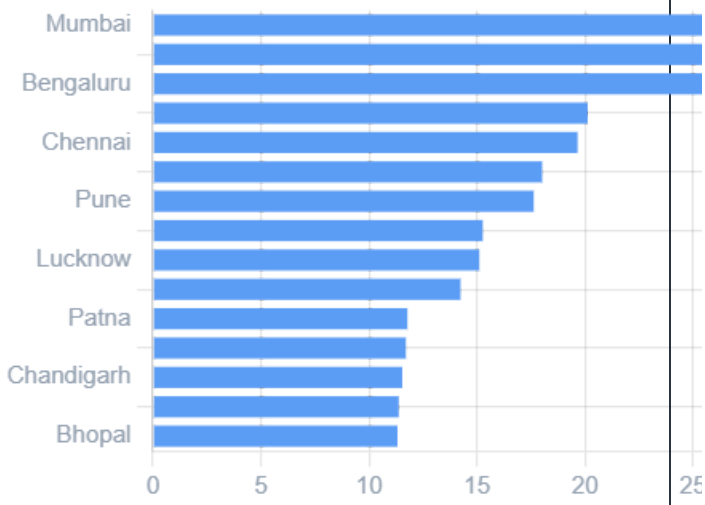
Spend by Category (₹ Cr)



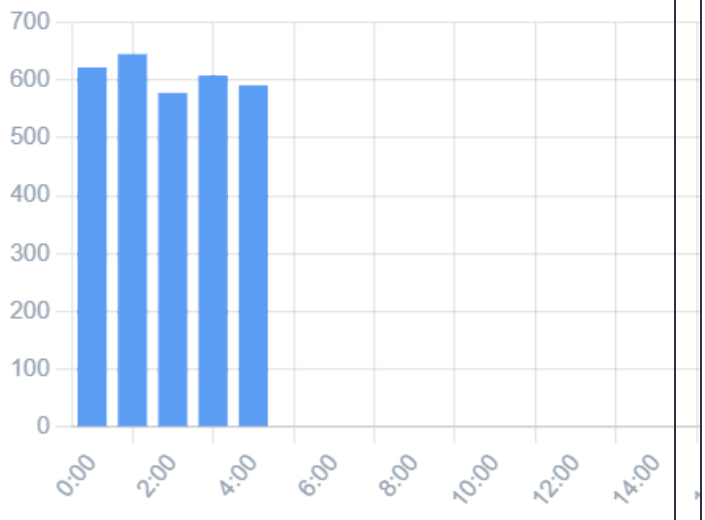
Transactions by Payment Mode



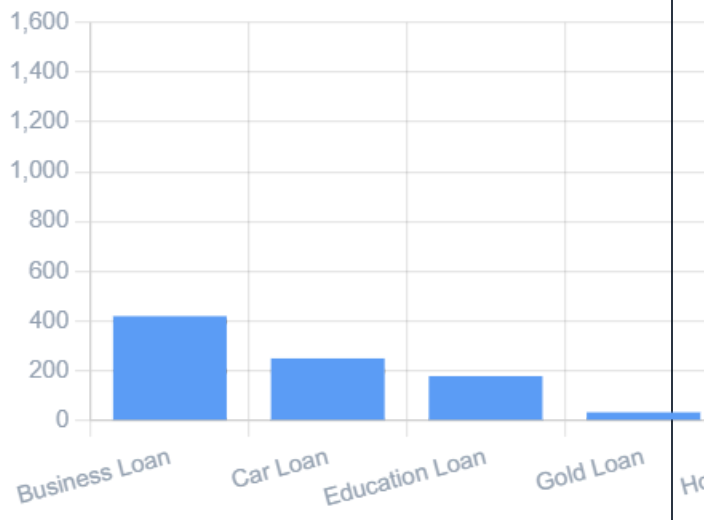
Spend by City (₹ Cr)



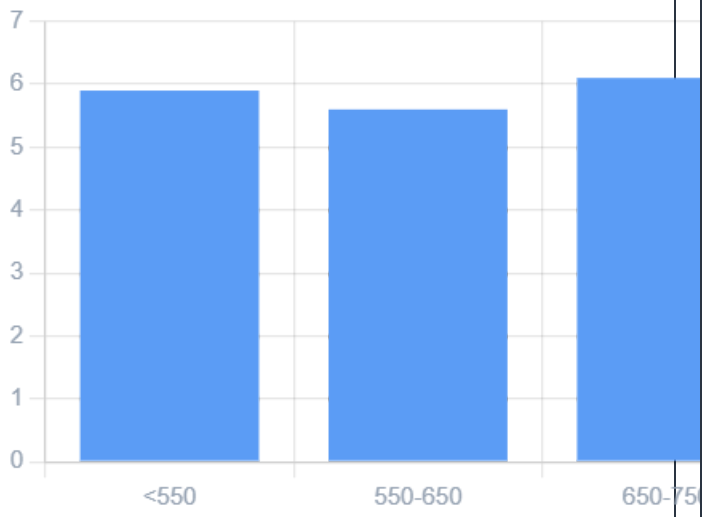
Fraud Transactions by Hour of Day



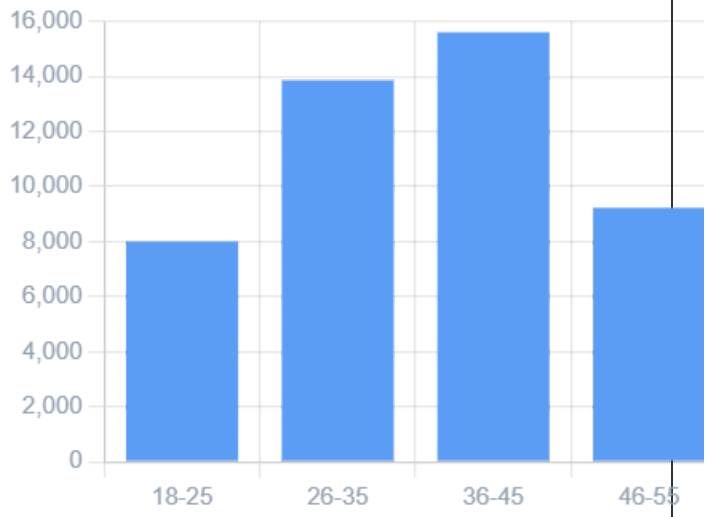
Loan Book by Type (₹ Cr)



Default Rate % by Credit Score Band



Customers by Age Band



Loan Portfolio by Type

Type	Loans	Amount (₹ Cr)	Default %	Avg Rate %
Business Loan	2,497	420.3	6.0	12.0
Car Loan	3,626	250.5	5.1	9.5
Education Loan	2,000	179.4	5.1	9.0
Gold Loan	1,966	34.0	6.0	9.5
Home Loan	4,004	1575.2	5.9	8.6
Personal Loan	5,907	200.8	6.3	13.0

Key Insights

Key insights

- **Spend is flat:** monthly debit spend stays around ₹7–8 Cr with no growth trend; the only dip is Feb (fewer days).
- **UPI dominates:** 177K of ~480K successful transactions (~37%), ahead of debit card and ATM. UPI volume is flat across 2023–2025.
- **Big-ticket categories:** EMI, Transfer, Rent and Investment together are ~60% of total debit spend.
- **Fraud:** 3,044 cases (0.61%) worth ~₹19.8 Cr. Average fraud ticket is ₹65K vs ₹10.5K for normal txns (about 6×). All fraud happens between 00:00 and 04:59, while only ~21% of normal txns happen then. UPI has the most fraud cases (1,132).
- **Credit risk:** default rate is 5.8% and overdue 6.0%; Personal (6.3%), Business and Gold (6.0% each) loans are the riskiest. Home loans are 59% of the book value (₹1,575 Cr of ₹2,660 Cr).
- **Customers:** 26–45 age group is 59% of customers; Business owners earn the most (₹1.04 lakh/month avg). Top 10% of customers account for 26.6% of spend.
- **Data quality:** 600 missing credit scores, 1,500 missing education, 800 missing marital status. No duplicate transaction IDs. 12% of accounts are Dormant.

Recommendations

- Add a rule-based fraud alert: block or step-up verification for high-value (>₹50K) UPI and card transactions between 00:00 and 05:00.
- Run a reactivation campaign for the 12% dormant accounts and 12% inactive customers.
- Tighten collections on Personal, Business and Gold loans (highest default) and monitor the ~1,200 overdue loans before they turn NPA.
- Push cross-sell (cards, investment, insurance) to the 26–45 age band and high-spend top 10% customers.
- Reduce the 3.5% failed-transaction rate by checking gateway and mode-wise failures.

Note: this is synthetic data. Credit score, income and spend were generated independently, so they show no correlation with default or spend. In real bank data, you would expect clear relationships there; use this as a practice dataset for method, not for business conclusions.